

06. Speech Synthesis



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Outline

6.1. Overview

6.2. Speech Synthesis Techniques

6.3. System Architecture

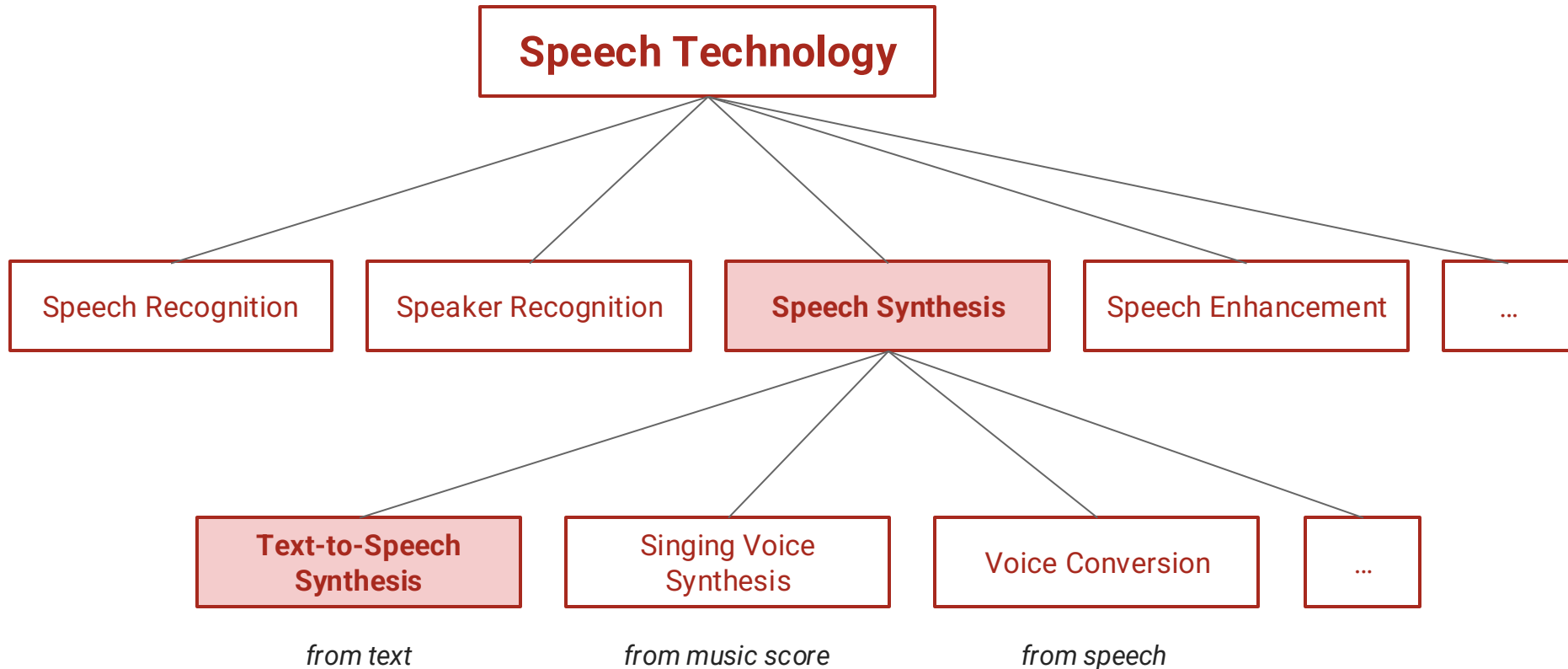
6.4. Challenges and Solutions

6.1 Overview

- ❖ Speech Synthesis - Tổng hợp tiếng nói
 - Quá trình nhân tạo sinh ra tiếng nói của con người
- ❖ Text-To-Speech (TTS) - Chuyển từ văn bản thành tiếng nói
 - Một loại tổng hợp tiếng nói
 - Quá trình chuyển đổi từ văn bản ngôn ngữ tự nhiên sang tiếng nói đầu ra

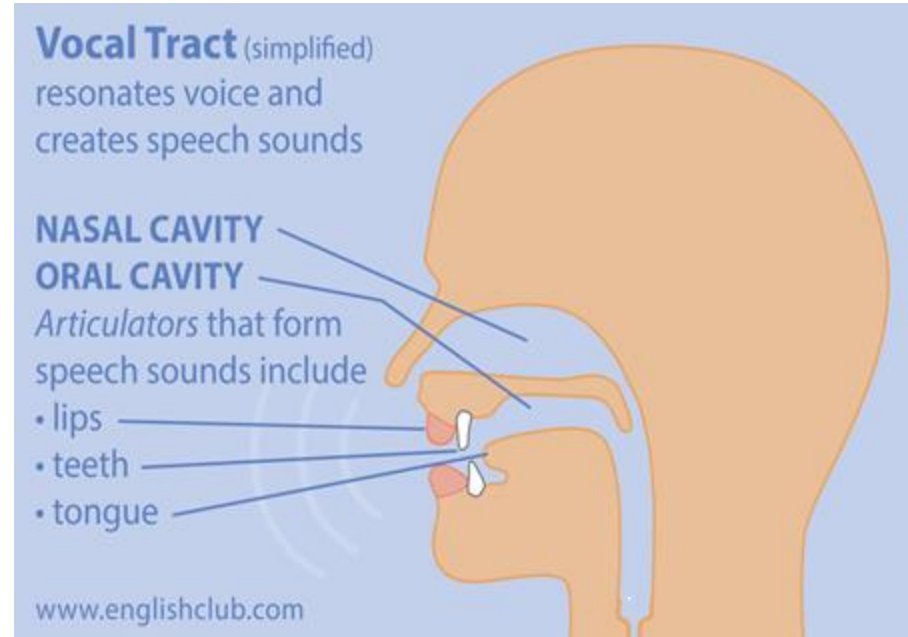
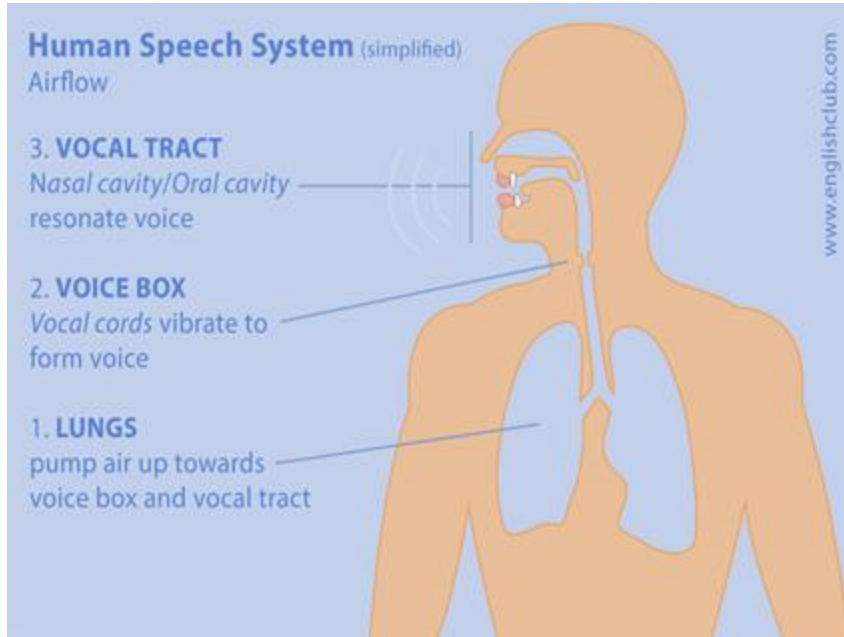


6.1. Overview



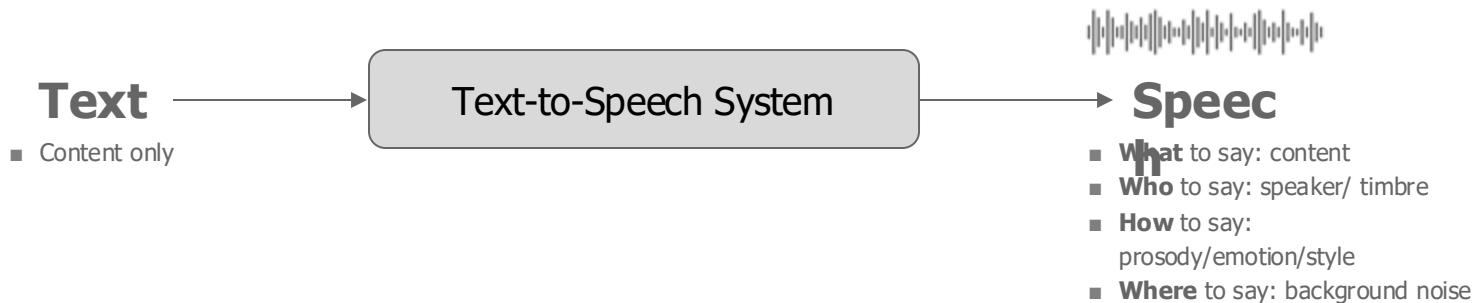
Text-to-Speech Synthesis

- The generation of human-like speech from text.
 - Human speech system

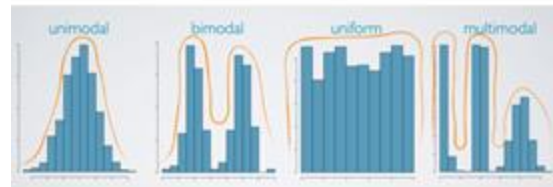


Text-to-Speech Synthesis

- The generation of **human-like speech** from text
 - Text-to-Speech (TTS) System



- Text-to-Speech Mapping is **One-to-Many**
 - Usually not **single-modal**, but **multi-modal**



History of Speech Synthesis

- 1003 - Máy móc bắt chước giọng nói đầu tiên ra đời (chế tạo bởi Gerbert ở Aurillac)
- 1779 - Mô hình bắt chước giọng nói người với năm nguyên âm ([a], [e], [I], [o] và [u])
- 1791 - Cải tiến tạo ra mô hình của lưỡi và môi, tạo ra phụ âm thêm vào nguyên âm
- 1857 - M.Faber chế tạo máy 'Euphonia'
- 1930 - Máy VOCODER, phân tích và tổng hợp giọng nói điều khiển bằng bàn phím, phát âm rõ ràng
- 1950 - Các hệ thống tổng hợp giọng nói bằng kỹ thuật điện tử ra đời
- 1961 - John Larry Kelly, Jr dùng máy tính IBM 704 để tổng hợp giọng nói
- Hiện nay - Hàng trăm hệ thống tổng hợp giọng nói, tiến hóa nhanh với chất lượng giọng nói đạt tới ngang người thật

Applications



1. Voice assistant



2. Content creation



3. Movie Dubbing



4. Text narration

Ứng dụng thực tế

- ❖ Hệ thống thông báo tự động
- ❖ Số hoá dữ liệu
 - Báo nói, sách nói
 - Số hoá bài giảng
 - Thuyết minh phim
 - Số hoá nội dung
- ❖ Giao tiếp bằng tiếng nói
 - Tổng đài tự động
 - Trợ lý ảo
 - Voicebot
 - Vạn vật kết nối
- ❖ Hỗ trợ người khuyết tật
 - Người khiếm thị
 - Người mất khả năng nói, diễn đạt...



6.2. Speech Synthesis Techniques

- **Articulatory Synthesis**

- Complicated, very low quality

- **Concatenative Synthesis**

- High intelligibility, but requires huge database, less natural and emotionless

- **Statistical Parametric Synthesis**

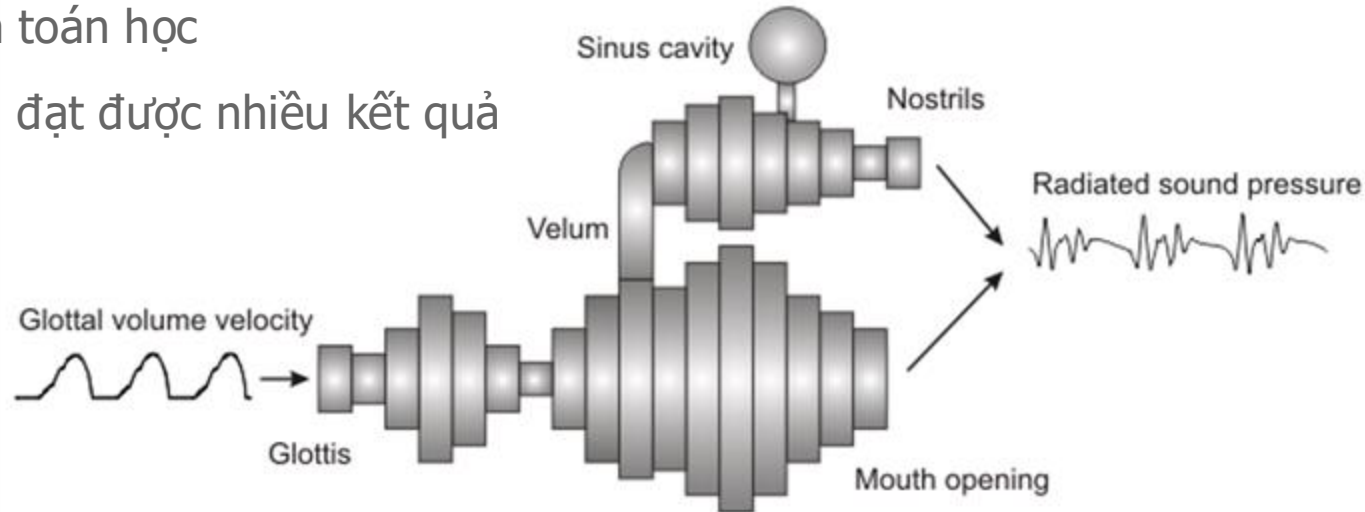
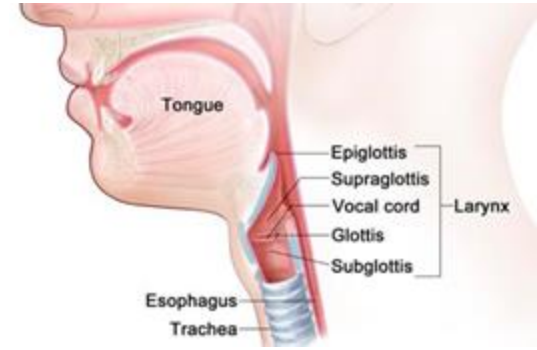
- Lower data cost and more flexible, but lower quality and robotic

- **Neural-based End-to-end Synthesis**

- Huge quality improvement, less human preprocessing and feature development.

Mô phỏng bộ phận cấu âm (1958)

- Replicate the movement of the human vocal tract to generate speech
- Mô phỏng các thành phần của bộ máy phát âm con người bằng các mô hình toán học
- Phức tạp → Chưa đạt được nhiều kết quả



Thảo luận

❖ Anh chị biết đơn vị âm (speech unit) nào?

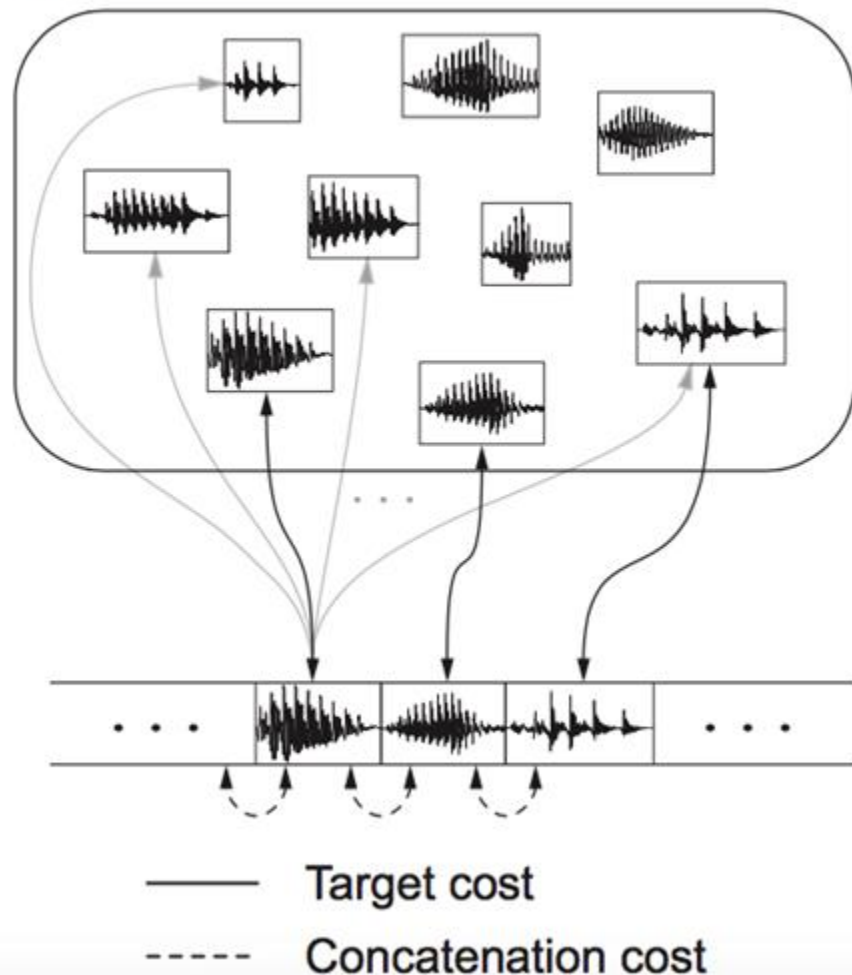
Thảo luận – Các đơn vị âm (speech unit)

- ❖ Câu nói – Utterance
- ❖ Đoạn tiếng nói – Prosodic Phrase
- ❖ Từ – Word
- ❖ Tiếng – Syllable
- ❖ Âm vị – Phoneme: Smallest perceptible segment

Concatenative Synthesis

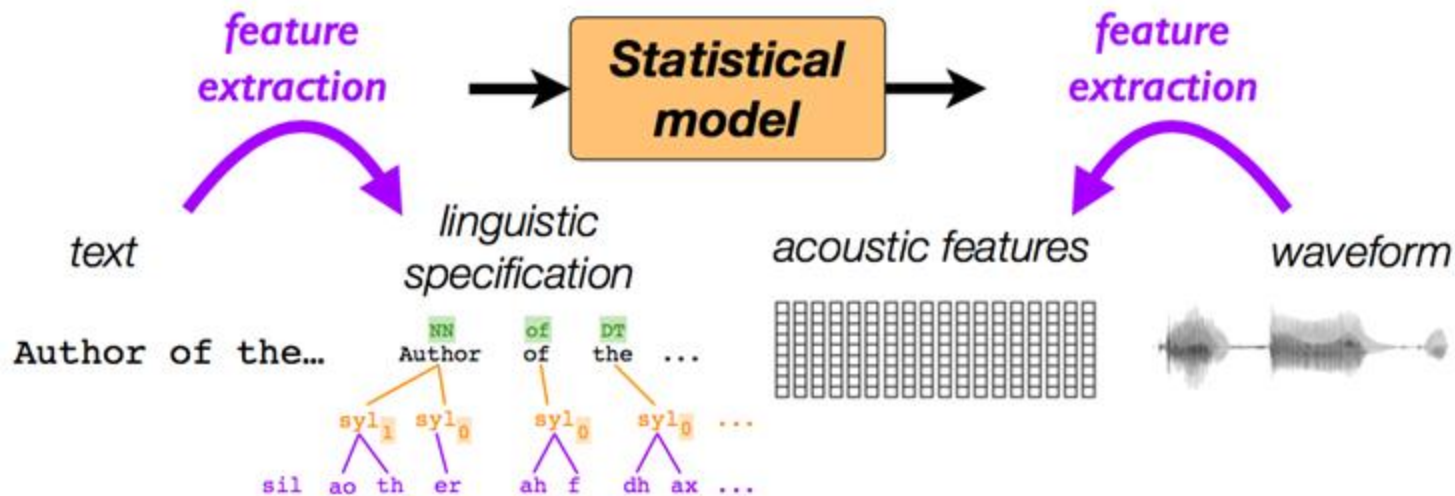
- Builds speech by concatenating segments of recorded speech (phones, syllables, or words).
- High naturalness, but limited flexibility
- Types:
 - Unit selection synthesis
 - Diphone synthesis

All segments

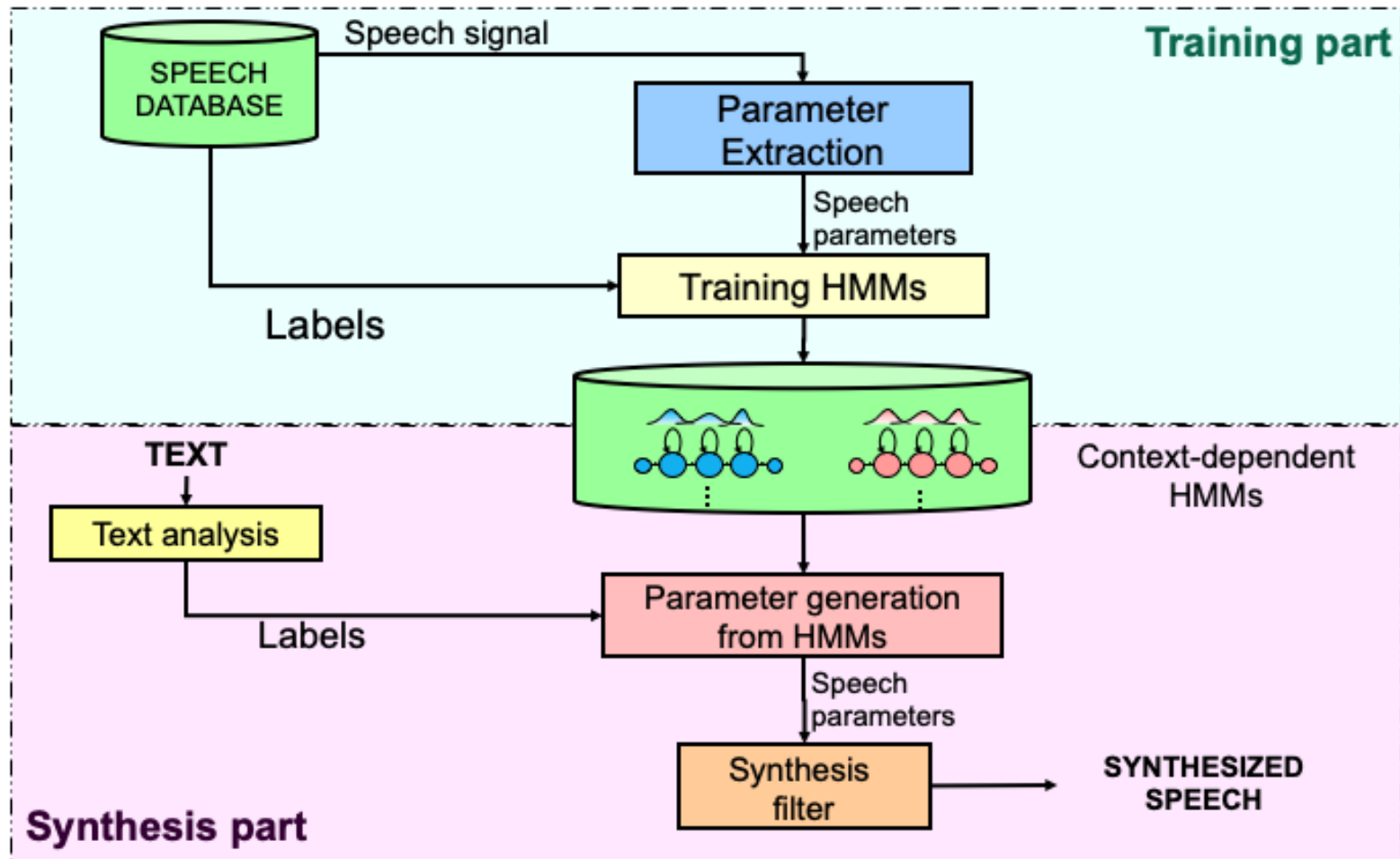


Statistical Parametric Synthesis

- Models speech acoustics and prosody statistically (e.g. HMMs)
 - Generating the average of some sets of similarly sounding speech segments (Zen 2009)
- Smaller footprint, more flexible, but less natural

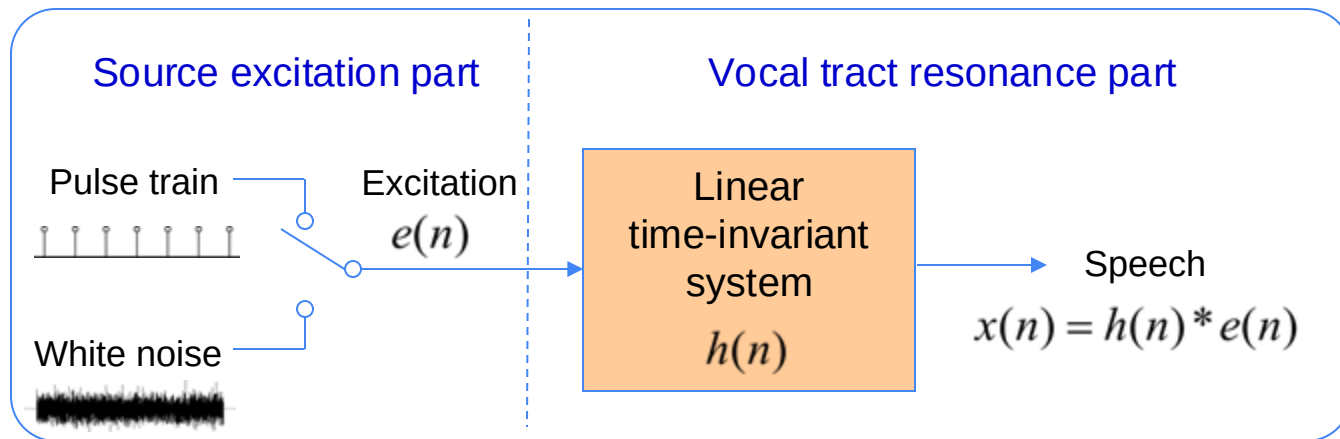


HMM
based
Speech
Syn-
thesis



Yoshimura (2002). Yoshimura Takayoshi. *Simultaneous Modeling of Phonetic and Prosodic Parameters, and Characteristic Conversion for HMM-Based Text-To-Speech Systems*. PhD thesis, Nagoya Institute of Technology, Japan, 2002.

Source-filter model (Vocoder)



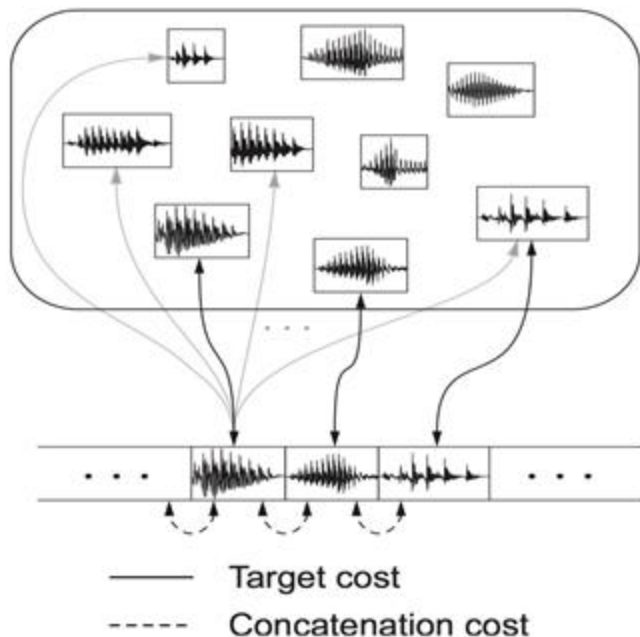
$$x(n) = h(n) * e(n)$$

↓ Fourier transform

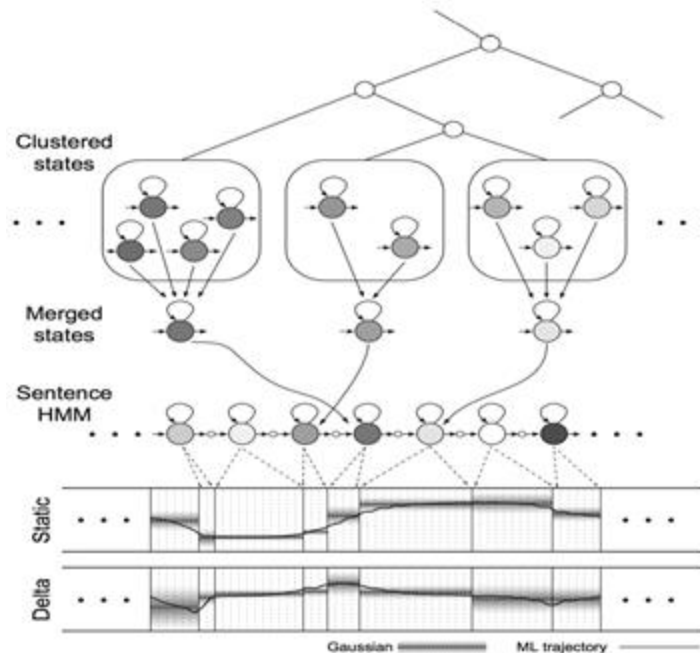
$$X(e^{j\omega}) = H(e^{j\omega})E(e^{j\omega})$$

$H(e^{j\omega})$ should be defined by the state-output vector of HMMs
(e.g., mel-cepstral coefficients, LSP coefficients)

Concatenation and Statistical Parametric Synthesis



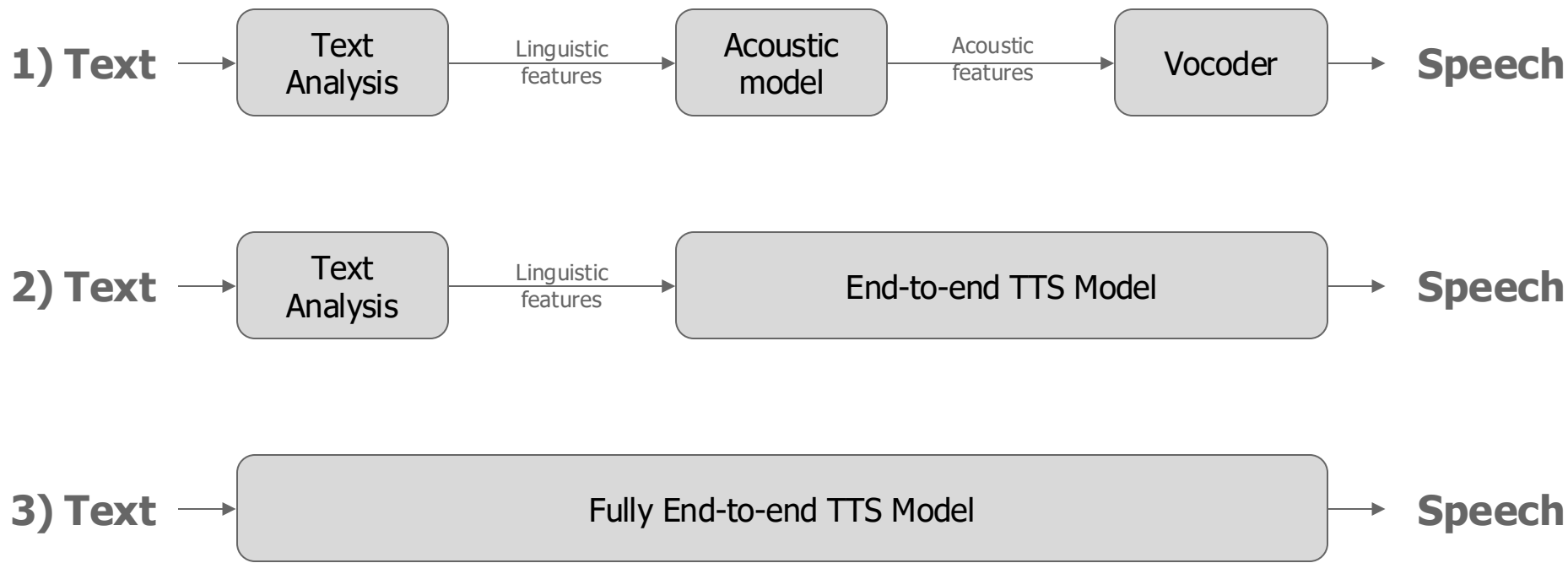
From **concatenation** to
unit selection synthesis



From **vocoding** to
statistical parametric synthesis

Zen (2009). Zen Heiga et al. Statistical parametric speech synthesis. Speech Communication.

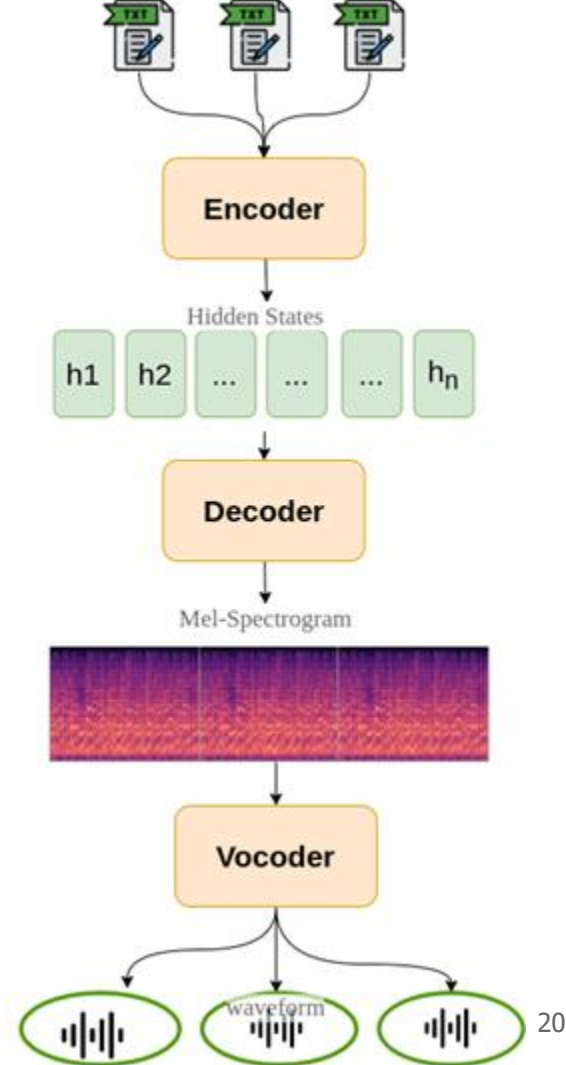
Neural-based Text-to-Speech



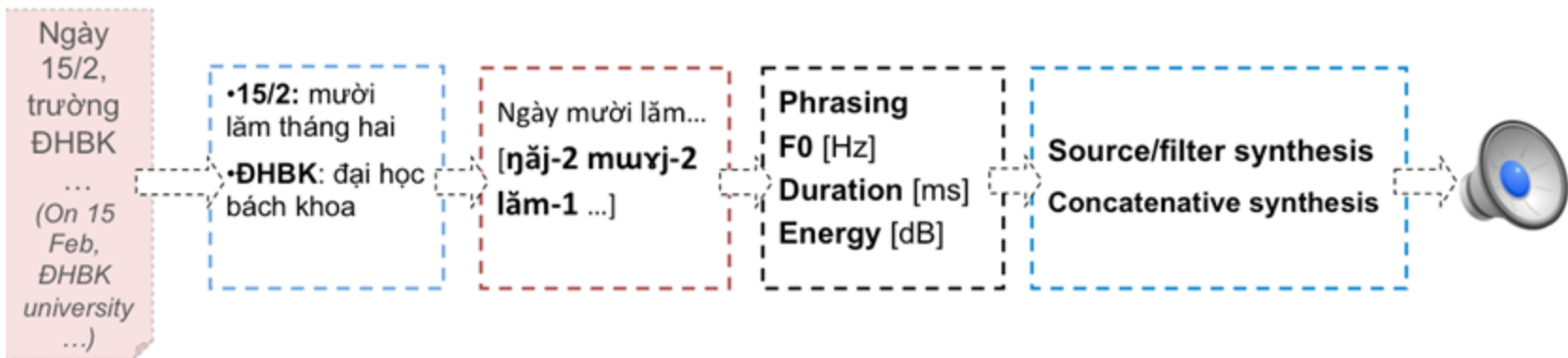
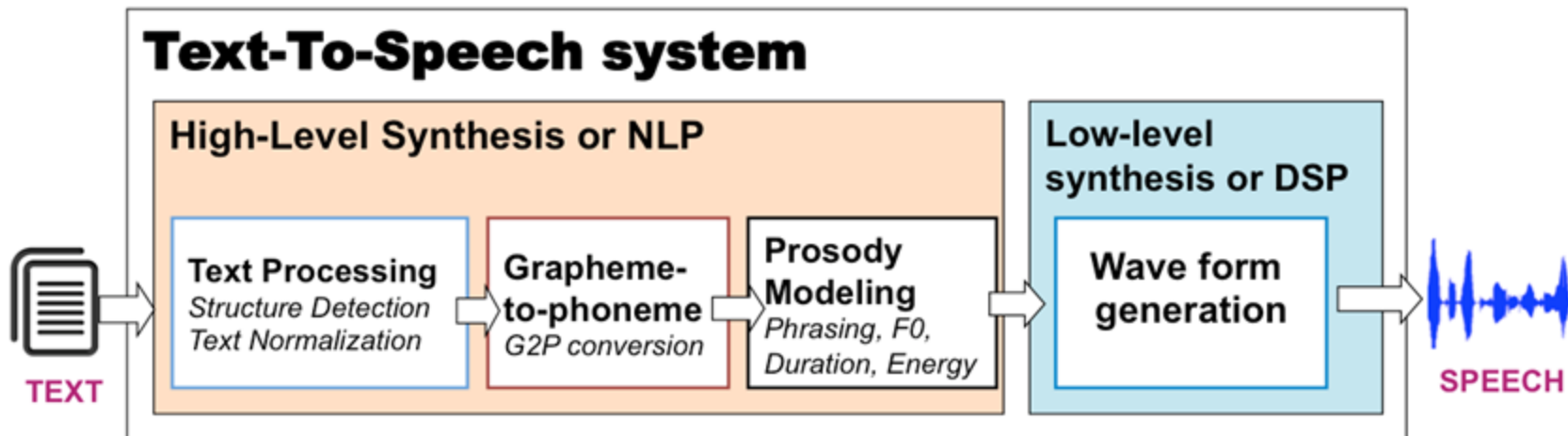
Towards a Fully end-to-end TTS model

Neural-based Speech Synthesis

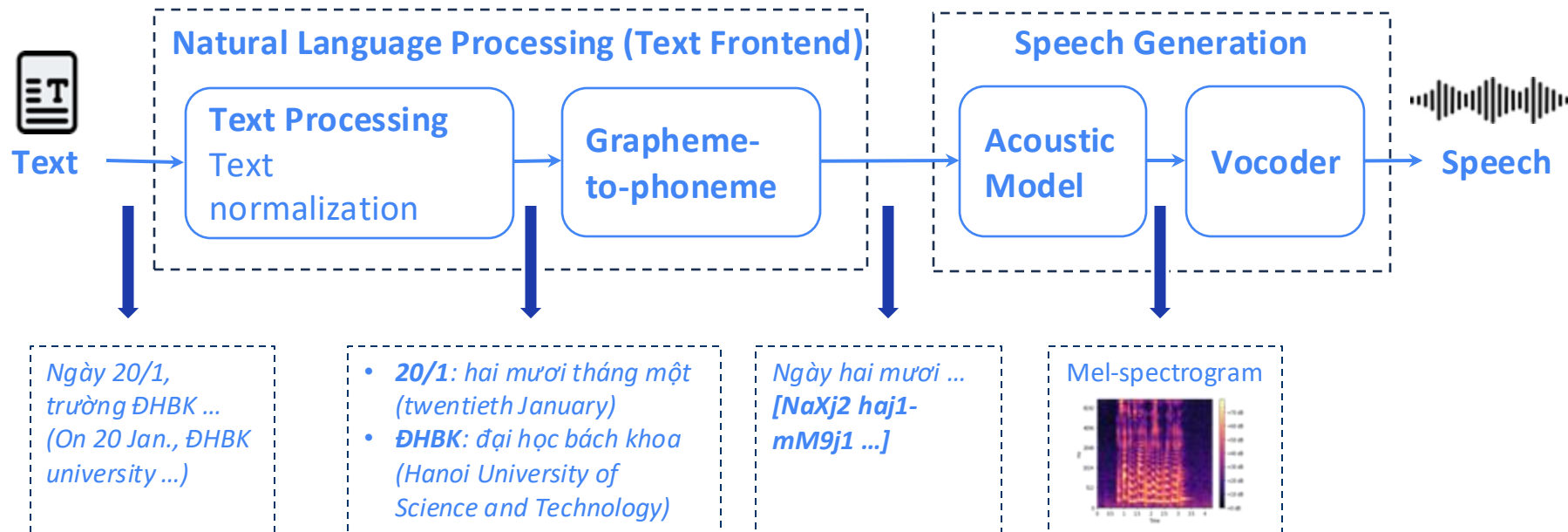
- **Encoder:**
 - Transforms the input text into a sequence of real-valued vectors and encodes them into hidden representations suitable for training deep neural networks.
- **Decoder:**
 - Decodes the hidden representations predicted by the neural network into acoustic features, typically in the form of mel-spectrograms.
- **Vocoder:**
 - Converts the mel-spectrogram into a time-domain audio waveform (speech signal).



6.3. TTS System Architecture - Traditional



6.3. TTS System Architecture - Modern

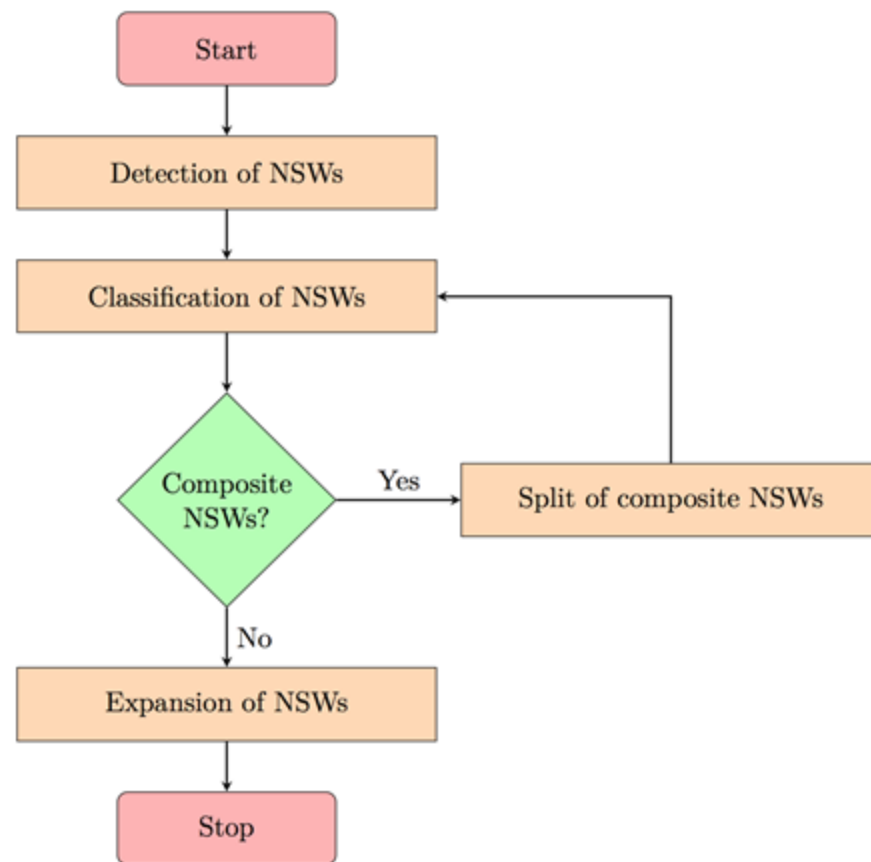


Text Analysis

- Transforms input text into linguistic features, including
 - **Text normalization:** *eg. Feb. 25th → "February twenty fifth"*
 - **Phrase/word/syllable segmentation:** *eg. "synthesis" → syn-the-sis*
 - **Part of speech (POS) tagging:** *eg. "I study at HUST" → [PRON, VERB, PREP, NOUN]*
 - **Grapheme-to-Phoneme conversion:** *eg. "speech" → s p i y ch*
 - ...
- Linguistic features can be:
 - Phoneme
 - Syllable, Word
 - Phrase and sentence-level features, ...

Text Normalization model

Group	Category	Description	Example
NUMBERS	NTIM	time	1:30
	NDAT	date	17/3/87, 1/3/2010
	NDAY	day and month	17/3, 03-05/3
	NMON	month and year	3/87, 3/2010, 3-5/87
	NNUM	number	2009, 70.000
	NTEL	telephone number	0915.334.577
	NDIG	number as digits	Mã số 999
	NSCR	score	Tỉ số là 3-5
	NRNG	range	Từ 3-5 ngày
	NPER	percentage	93%, 30-40%,
	NFRC	fraction	34/6, 45,6/145
	NADD	address	số 14/3/2 phố Huế
LETTERS	LWRD	read as a word	London, NATO
	LSEQ	letter sequence	ODA, GDP
	LABB	abbreviation	TS (tiền sĩ)
OTHERS	PUNC	speakeable punctuation	... () [] ‘ ’ “ ” - /
	URLE	url, path name or email	http://soict.hut.vn
	MONEY	Money	2\$, \$2, VNĐ 9.000
	CSEQ	read all characters	:), XXX
	DURA	duration (nghi)	“-” in scores (2-3)
	NONE	ignored	ascii art...



Vietnamese word segmentation

- Same delimiter (spaces) for Vietnamese syllables and words
 - E.g. Tôi | đang | nghiên cứu | về | tổng hợp | tiếng nói | Tiếng Việt | tại | phòng thí nghiệm | LIMSI (I am doing research on Vietnamese speech synthesis in the LIMSI laboratory).
- Ambiguity in Vietnamese word segmentation
 - E.g. Học sinh | học sinh | học OR Học sinh | học | sinh học
(Pupils, pupils study OR Pupils study biology)

Vietnamese Word Segmentation Tools

Tool	Provider	Programming Language	Technique	Strengths	Notes
VnCoreNLP (RDRSegmenter)	VNU-HCM (Vietnam National University)	Java	Rule-based (Ripple Down Rules)	High accuracy, robust, widely used	Core of most Vietnamese NLP pipelines
pyvi	Open-source community	Python	CRF-based	Lightweight, Pythonic	Good for quick prototyping
underthesea	UnderTheSea team (Vietnam)	Python	Hybrid (rules + statistical)	Easy to use, pip-installable	Lower accuracy than RDRSegmenter
VnTokenizer	Le Hong Phuong (HCMUS)	Java	Rule-based	Historical significance	Less used in modern applications

Building Vietnamese tonophone set

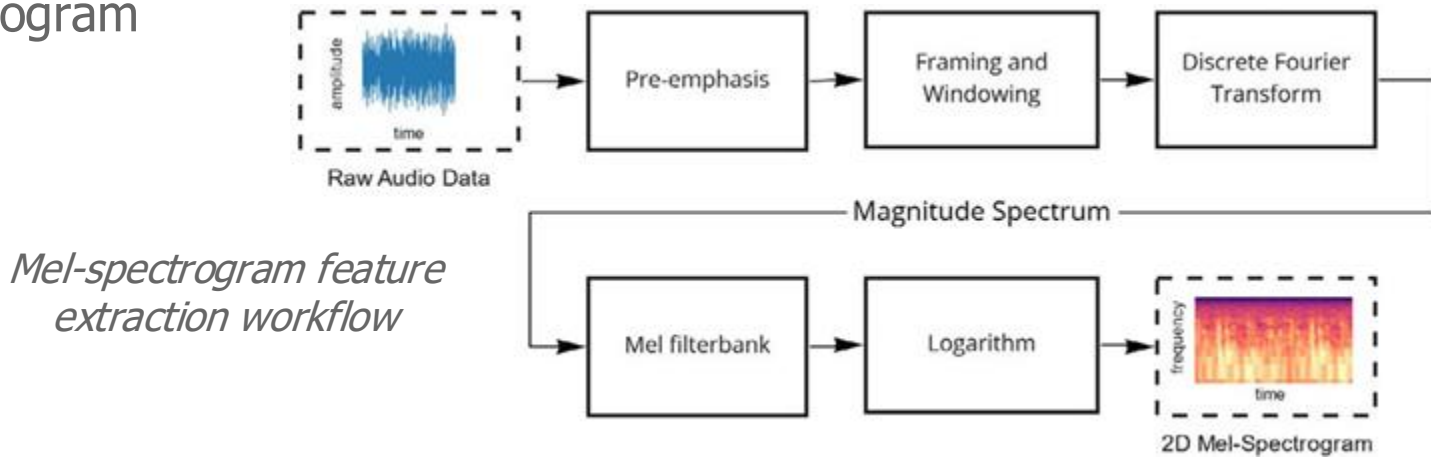
- **Tonophone**: An allophone with tone information
 - Adding tones to all elements of rhyme
 - E.g. chuyển [tɕwiən-3] (to move) => [tɕ], [w3], [iə3], [n3]

Element	Allophones	Lexical tones	TN #
Initial consonant	ɲ t ^h tɕ ɲ f X z ɣ d t k ɓ m n l s h v p	(NOT adding)	19x1
Medial	w	1-4, 5a, 5b, 6a, 6b	1x8
Nucleus	i e ɛ a ɤ ɯ u o ɔ ɛ̃ ǎ ǎ̃ ǎ̃̃ iə ʊə uə	1-4, 5a, 5b, 6a, 6b	16x8
Final consonant	m n ɲ ɲm ɲ. w j	1-4, 5a, 6a	7x6
	p t k kp k̚	5b, 6b	5x2
TOTAL	48		207

- **Inventory** of all vietnamese pronounceable syllables
 - PRO-SYLDIC (PROnounceable SYLLable DICTIONary)
 - Maintaining non-existent but pronounceable syllables for loan words
 - For automatic transcription and text normalization

Acoustic model

- **Predicts** acoustics features from linguistic features.
- **Aligns** linguistic and acoustic features
 - Maps text-based sequences (eg. phonemes) to continuous speech representations
- **Acoustic features** can be:
 - Mel-scale Frequency Cepstral Coefficients (MFCC)
 - Mel-spectrogram



Acoustic Models (Text → Spectrogram)

Type	Description	Representative Models	Pros	Cons
Autoregressive	Predict spectrogram frame-by-frame with attention	Tacotron, Tacotron 2	Natural prosody	Slow, unstable attention
Non-Autoregressive	Parallel prediction using duration predictor	FastSpeech, FastPitch, FastSpeech 2	Fast, stable	Requires alignment
Variational / Flow-based	Latent modeling with VAE / Normalizing Flow	Glow-TTS, Flow-TTS	Expressive, flexible	Complex training
Diffusion-based	Generate spectrogram via denoising process	Grad-TTS, FastDiff, DiffGAN-TTS	High-quality, controllable	Slow inference

Acoustic Models

Mô hình	Kiến trúc	Autoregressive	Alignment	Tốc độ sinh (Inference)	Đặc điểm nổi bật
Tacotron 2	Seq2Seq + Attention	✅ Có	Attention (soft)	❌ Chậm (tuần tự)	Dễ lỗi lặp/thiếu từ; phổ biến
FastSpeech	Transformer + Duration	❌ Không	Dựa trên teacher-forced duration từ Tacotron	✅ Nhanh (song song)	Ổn định, nhanh hơn Tacotron
FastSpeech 2	Transformer + Duration	❌ Không	Dựa trên duration (tự học hoặc teacher)	✅ Rất nhanh	Thêm pitch & energy modeling
Glow-TTS	Flow-based (Glow)	❌ Không	Monotonic Alignment Search (MAS)	✅ Rất nhanh	Không cần attention; sinh mel theo flow
VITS	VAE + Flow + GAN	❌ Không	Variational + Duration latent align	✅ Nhanh + Real-time	End-to-end sinh waveform luôn

Acoustic model - Example

- **Glow-TTS**

- Log likelihood

$$\log P_X(x|c) = \log P_Z(z|c) + \log \left| \det \frac{\partial f_{dec}^{-1}(x)}{\partial x} \right|$$

- Prior distribution is learnt from phoneme text

$$\log P_Z(z|c; \theta, A) = \sum_{j=1}^{T_{mel}} \log \mathcal{N}(z_j; \mu_{A(j)}, \sigma_{A(j)}),$$

- Alignment matrix A is obtained by monotonic alignment search

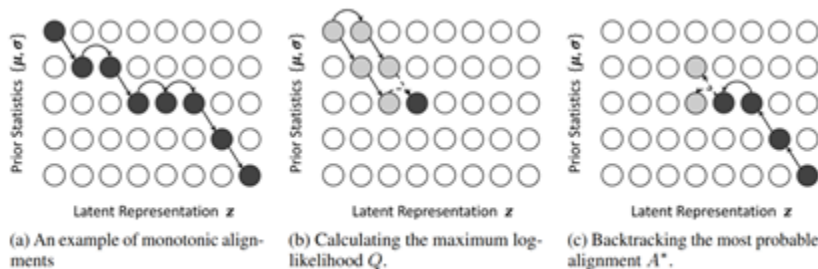
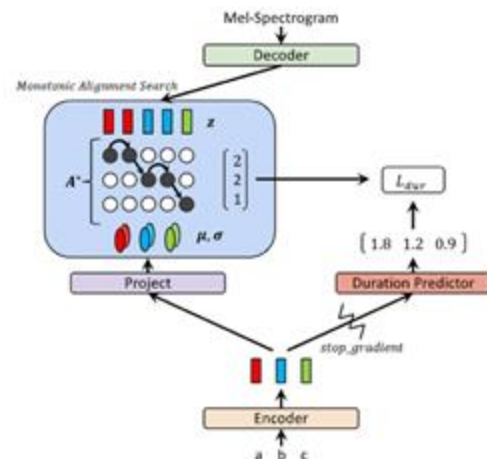
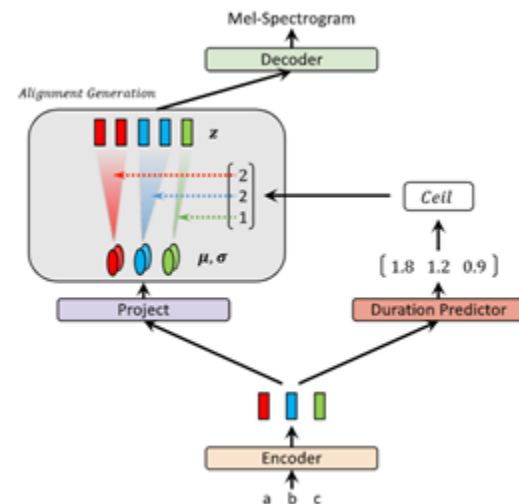


Figure 2: Illustrations of the monotonic alignment search.



(a) An abstract diagram of the training procedure.



(b) An abstract diagram of the inference procedure.

Vocoder

- **Voice encoder**
 - **Final stage in TTS pipelines, transforming** acoustics features **(typical spectrograms) into intelligible, natural-sounding speech.**
- Greatly impacts the **naturalness, quality, and intelligibility** of synthesized speech
 - Traditional vocoders like Griffin-Lim are simple but produce robotic speech, while neural vocoders such as HiFi-GAN offer high-quality, real-time synthesis.
 - Vocoder architectures have evolved from autoregressive models (WaveNet) to more efficient GAN- and diffusion-based approaches

Vocoder (Spectrogram → Waveform)

Type	Description	Representative Models	Pros	Cons
Autoregressive	Frame-by-frame waveform generation	WaveNet	Very high quality	Very slow
Flow-based	Invertible transformation from noise → waveform	WaveGlow, FloWaveNet	Faster than AR	Quality not as good as GANs
GAN-based	Adversarial learning to generate waveform	MelGAN, HiFi-GAN, UnivNet	Fast, high-quality	Needs careful training
Diffusion-based	Denoising process to generate waveform	DiffWave, WaveGrad	High fidelity	Slower inference

Vocoder - Key challenges

- **Trade-off** between quality and speed
 - Autoregressive models are slow but high-quality.
 - Non-autoregressive models are fast in inference but resource-intensive during training.
- **Generalization** Across Speakers & Datasets
 - Vocoder trained on one dataset may struggle with unseen voices

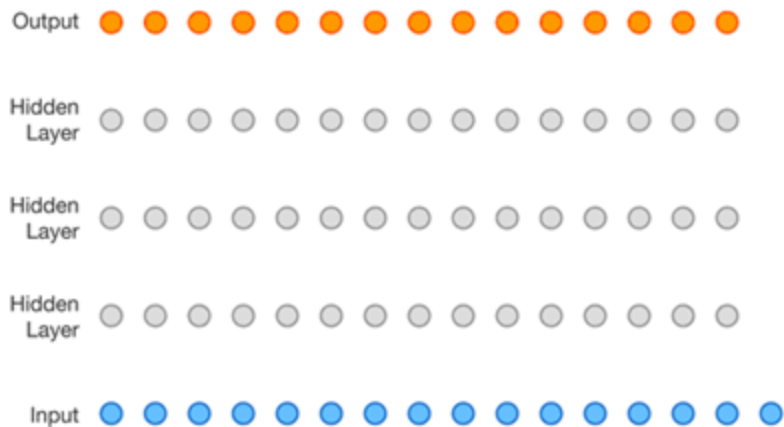
End-to-end TTS model

- **Advantages**

- Trained with text-speech pairs with minimum human annotation
- Do not require explicit alignment between text and speech
- Errors cannot accumulate and no error propagation since it is a single model

- **More end-to-end TTS models**

- Directly predict waveform instead of intermediate representation (eg. mel-spectrogram)



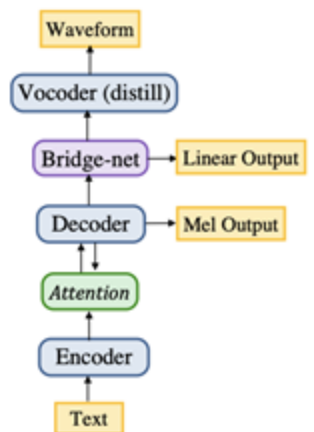
WaveNet: autoregressive model with dilated causal convolution

End-to-end TTS model

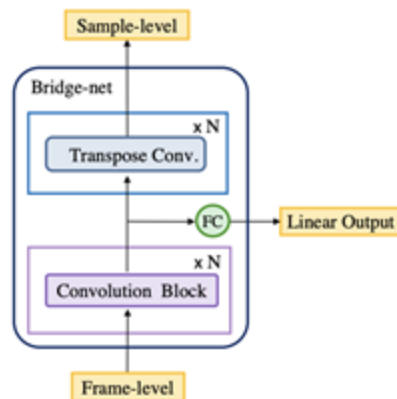
- **Advantages**

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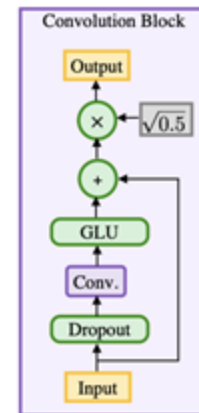
- **Fully en**



(a) Text-to-wave architecture



(b) Bridge-net



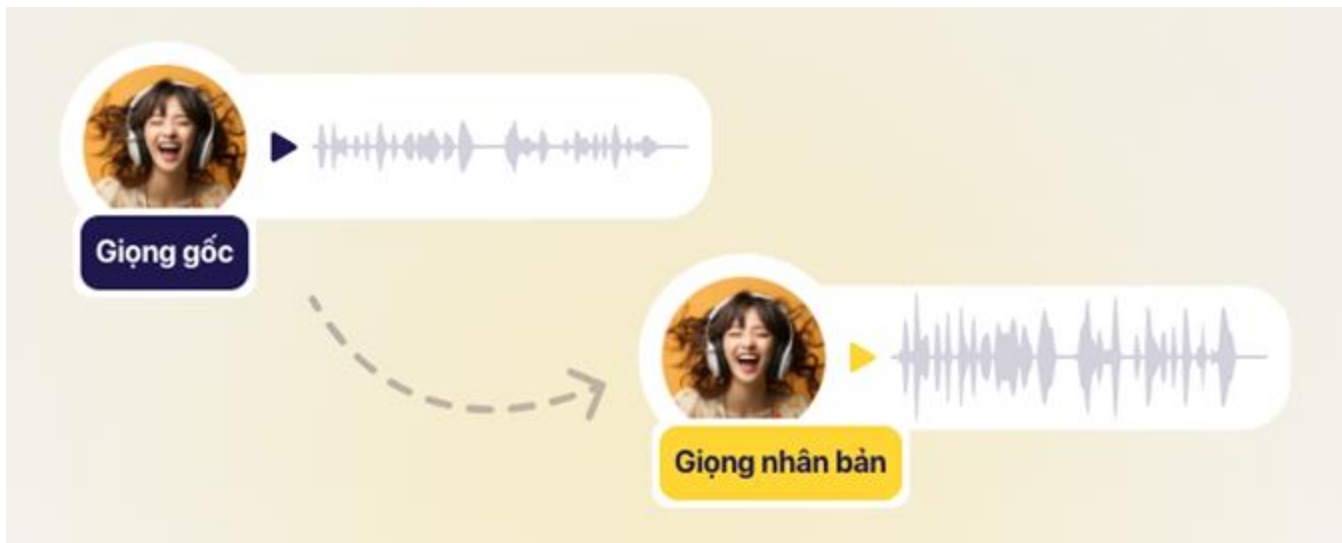
(c) Convolution block

End-to-End TTS Architecture (Text → Waveform)

Model Type	Description	Representative Models	Pros	Cons
VAE + Flow + GAN	Combines latent variables, flow-based decoder, and adversarial learning	VITS, VITS2	High-quality, compact, expressive	Complex training, harder to debug
Diffusion-based E2E	Text-to-waveform via diffusion	StyleTTS2, NaturalSpeech 2	Natural, controllable	Training/inference cost

Voice Cloning

Synthesizing speech that mimics the **voice identity** of a specific speaker — typically from limited data — while allowing arbitrary text as input.



TTS vs. Voice Cloning

Aspect	TTS (Text-to-Speech) 🗣️	Voice Cloning 🗣️👤
Definition	Converts text into synthetic speech.	Reproduces a specific person's voice.
Purpose	Generate natural, intelligible speech for general use.	Generate speech that sounds like a specific speaker .
Training Data	Large datasets from multiple speakers .	Requires data from one or a few target speakers .
Customization	Uses pre-trained voices , limited speaker selection.	Can replicate any speaker's voice.
Data Required	100+ hours (traditional), 1–10 hours (modern).	Few-shot learning (minutes to hours) for cloning.
Output Quality	Highly natural , but generic voice .	Highly realistic , captures unique speaker traits .
Speaker Consistency	Voice remains generic across all outputs.	Mimics speaker's tone, style, and pitch .

Voice Cloning Challenges

- Low-resource data:
 - Only a few minutes or samples of the target speaker's voice available
- Speaker identity loss:
 - Generated voice lacks vocal uniqueness.
- Flat prosody:
 - Robotic or emotionless delivery.
- Multilingual/tonal issues:
 - Difficulties with tone, pitch, or phoneme disambiguation (e.g., in Vietnamese).
- Evaluation:
 - Subjective and hard to automate.

Voice Cloning Models

- **Multi-speaker TTS + Embedding**

- Base: Tacotron 2, FastSpeech 2, VITS
- Speaker embedding is concatenated at encoder or decoder level.

- **End-to-End Voice Cloning**

- Text + reference audio → waveform
- No explicit mel-spectrogram stage; trained jointly (e.g. YourTTS, VITS2, StyleTTS2)

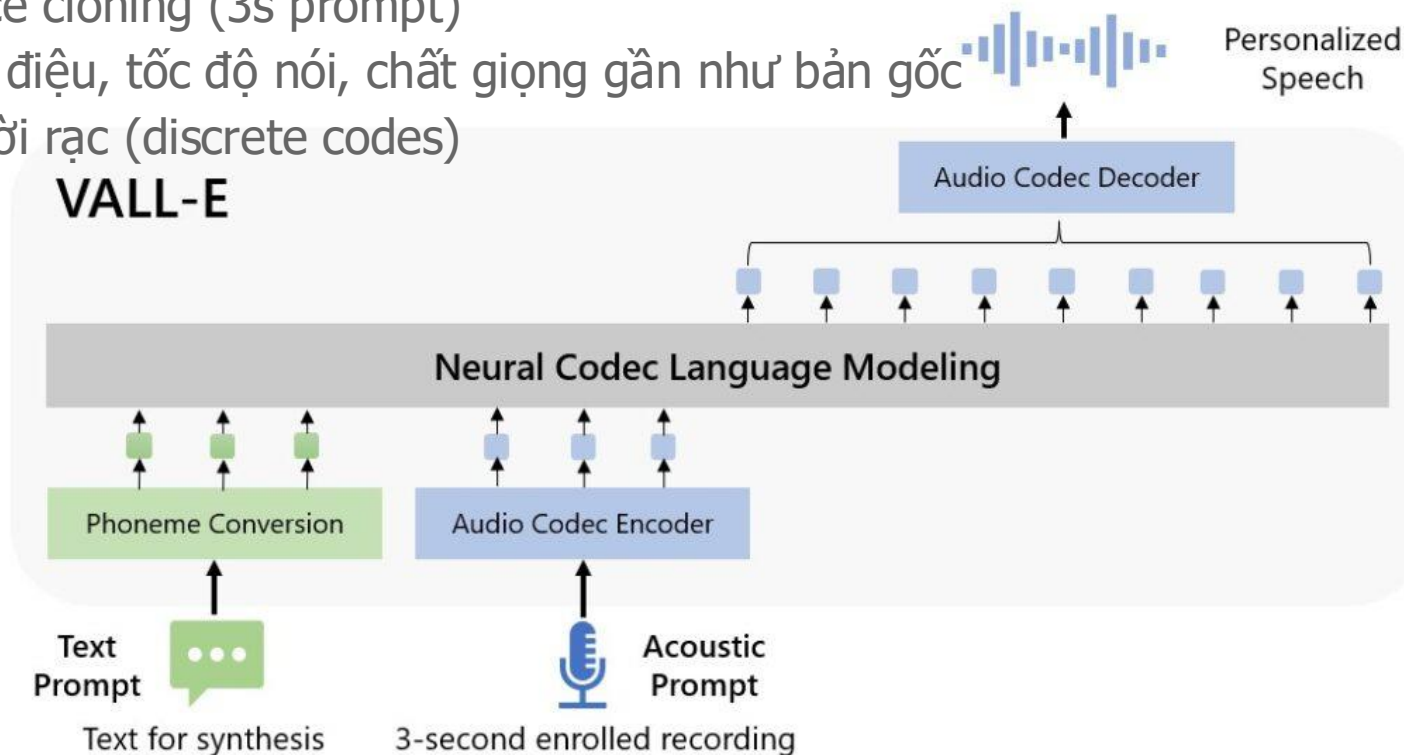
- **Zero-shot Cloning**

- Pretrained speaker encoder (e.g., GE2E) + shared TTS backbone
- Works with unseen speakers without fine-tuning (e.g. YourTTS, SV2TTS, hoặc iSTFT-VITS, VALL-E, StyleTTS2)

VALL-E: Neural Codec Language Model

- Microsoft Research (2023)
- Zero-shot voice cloning (3s prompt)
- Giữ được ngữ điệu, tốc độ nói, chất giọng gần như bản gốc
- Sử dụng mã rời rạc (discrete codes)

Không sinh spectrogram mà dự đoán mã RVQ rồi tái tạo



Các hướng nghiên cứu cải tiến VALL-E

- Prosody Modeling, Style Token, Emotion Modeling... để kiểm soát ngữ điệu, cảm xúc, tốc độ
- Giảm sự phụ thuộc vào prompt: Robust Speaker embedding, Speaker prompt từ text (zero-prompt), Speaker encoder
- Multilingual và cross-lingual voice cloning
- Mô hình hóa tốt hơn giữa text → code:
 - Semi-autoregressive/Diffusion model thay vì AR transformer → giảm lỗi lặp, mất đoạn
 - Tăng độ chính xác trong mapping giữa ngữ nghĩa câu và chuỗi code RVQ
- Tách biệt content / speaker / prosody: Speaker embedding (giọng), Content token (ngữ nghĩa), Style/prosody token
- Kết hợp với Large Language Models (LLMs):
 - LLM sinh ra prosody tokens, emotion hint, hoặc text chuẩn hóa trước
 - Dùng prompt text dạng ngữ cảnh: "nói với giọng nghiêm túc, trầm ấm..." → feed vào decoder

RVQ-VAE

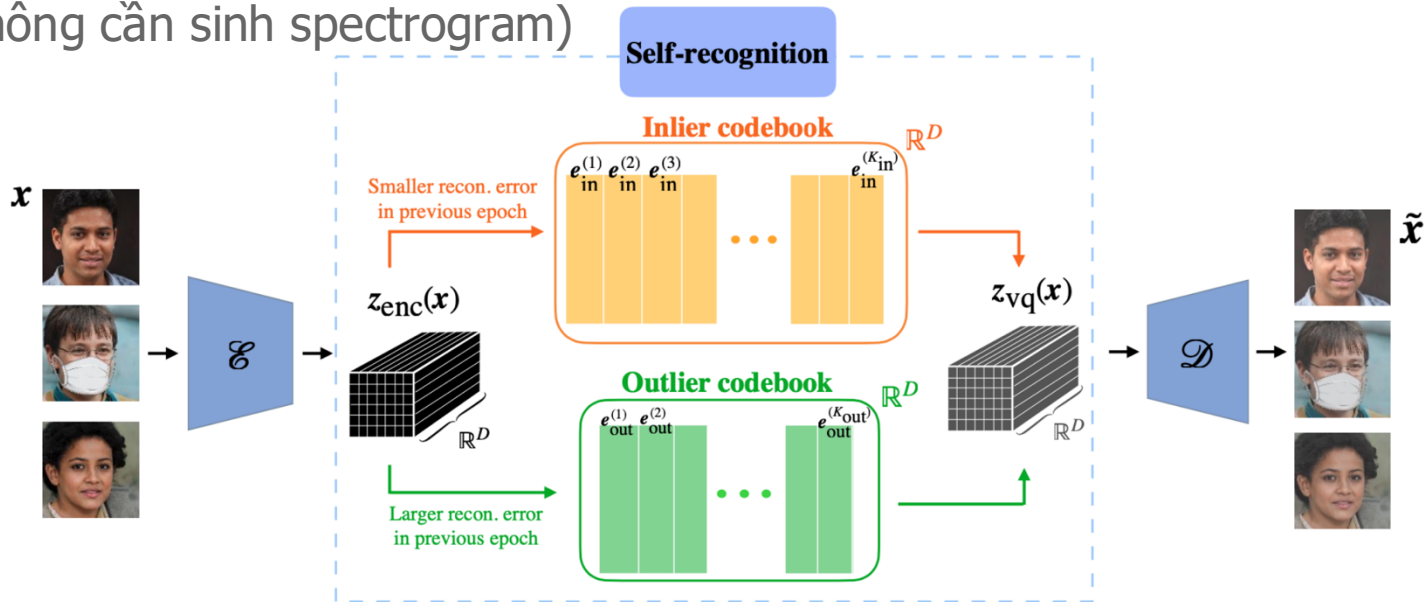
- **VAE** (Variational Autoencoder): học phân phối ẩn (latent distribution) để **nén và tái tạo dữ liệu**.
- **VQ-VAE** (Vector Quantized VAE): dùng **vector lượng tử hóa rời rạc (codebook)** để đại diện cho không gian ẩn.
- **RVQ (Residual Vector Quantization)**: cải tiến VQ bằng cách **sử dụng nhiều tầng lượng tử hóa kế tiếp**, mỗi tầng học phần dư (residual) còn lại từ tầng trước → Tăng khả năng biểu diễn mà không cần codebook quá lớn.

👉 **RVQ-VAE** = Autoencoder + multi-stage vector quantization

→ Học cách **mã hóa giọng nói thành chuỗi mã rời rạc giàu thông tin**, giúp mô hình dự đoán và tái tạo giọng nói linh hoạt và tự nhiên hơn, đồng thời hỗ trợ tốt cho **voice cloning, zero-shot TTS, và multilingual synthesis**.

Vai trò của RVQ-VAE với TTS

- Học latent speech representation: dùng làm mục tiêu huấn luyện
- Zero-shot speaker modeling: clone giọng từ mã speech ngắn
- Tách biệt giữa nội dung (content) và phong cách giọng nói (style)
- Giảm gánh nặng huấn luyện vocoder: chỉ cần học mapping từ text $\rightarrow$ mã RVQ (không cần sinh spectrogram)



AR Transformer trong TTS+RVQ

- Autoregressive decoding: Mỗi token dự đoán dựa trên toàn bộ các token trước đó
 - Cực kỳ chính xác về chất lượng giọng nói: Mô hình giữ được prosody, nhấn nhá, ngữ điệu tốt hơn
 - Chậm (không phù hợp với real-time): Vì phải sinh từng token một (không song song được)
- Phù hợp với offline high-quality TTS (giống như VALL-E), có thể dùng trong huấn luyện, sinh sample chất lượng cao làm "ground truth"
- Thường được dùng làm teacher model để tạo ground truth cho training NAR model (distillation)

NAR Transformer trong TTS+RVQ

- Sinh toàn bộ chuỗi mã cùng lúc hoặc song song theo nhóm Khả năng song song hóa cao
- Nhanh hơn nhiều lần so với AR: Rất phù hợp cho ứng dụng real-time TTS
- Khó giữ được nhịp điệu, intonation như AR: Dễ bị mất ngữ cảnh dài, đặc biệt nếu không có trợ lực (e.g. từ encoder)

→ Dùng để **deploy** online, hoặc dùng như student model trong distillation

→ Với sự kết hợp với RVQ, giúp giảm độ phức tạp của output (chỉ cần sinh mã discrete) nên NAR hoạt động hiệu quả hơn so với sinh mel trực tiếp.

Challenges in TTS developments

- **Inference Speedup**

- Reducing inference latency while maintaining high quality

- **Robust and Expressive TTS**

- Generating natural, expressive speech with accurate intonation, rhythm and emotion.

- **Low resource language**

- Achieving high-quality synthesis with minimal training data for low-resource languages.

Challenges and Solutions for Vietnamese TTS

- **Word Segmentation & Text Normalization**
 - Hybrid models
- **Lack of High-Quality Speech Data**
 - Semi-supervised or self-supervised learning
 - Data augmentation
- **Prosody Modeling**
 - Style, emotion tags
 - Style transfer or reference encoder

Summary

- TTS technology has **evolved** from **concatenative synthesis** and **statistical parametric synthesis** to **neural-based Text-to-Speech**, with modern approaches increasingly focusing on leveraging generative models.
- Mainstream TTS models typically use separate **acoustic models** and **vocoders**, but **fully end-to-end** models are emerging.
- The primary goal of TTS is to **improve quality** while **reducing costs**.